

Variants of ANNs

An Activation Function

- It transforms a neuron's net input signal into a single output signal to be broadcasted further in the network

A Network Topology (Architecture)

- It describes the number of neurons in the model as well as the number of layers and manner in which they are connected.

The Training Algorithm

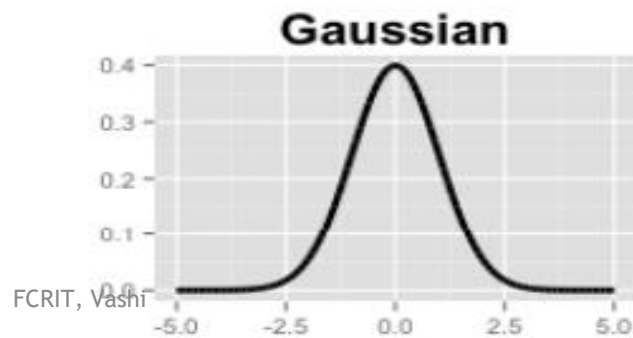
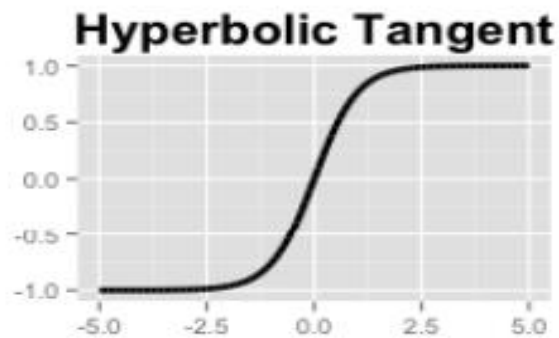
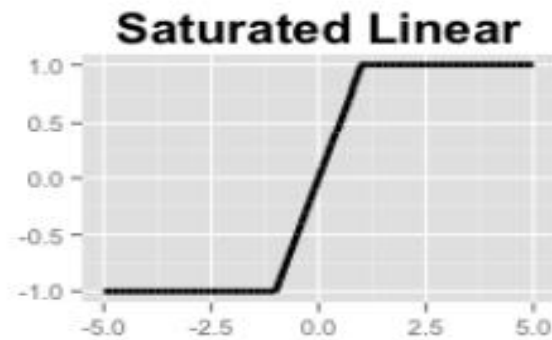
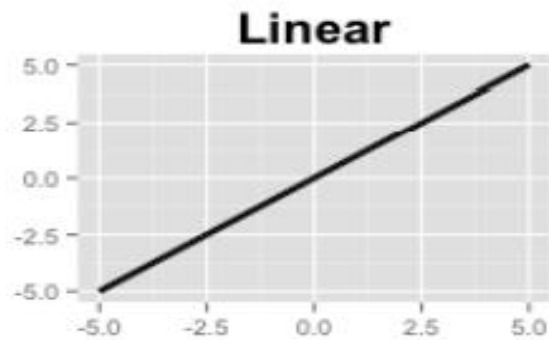
- It specifies how connection weights are set in order to inhibit or excite neurons in proportion to the input signal.

Activation Function

- ▶ The activation function is the mechanism by which the artificial neuron processes information and passes it throughout the network.
- ▶ The process involves summing the total input signal and determining whether it meets the firing threshold.
- ▶ If so, the neuron passes on the signal; otherwise, it does nothing.
- ▶ In ANN terms, this is known as a threshold activation function, as it results in an output signal only once a specified input threshold has been attained.

Activation Function

- Some neural network algorithms allow a choice of alternatives. A selection of such activation functions is as shown:



Network Topology

- ▶ The capacity of a neural network to learn is rooted in its topology, or the patterns and structures of interconnected neurons.
- ▶ Although there are countless forms of network architecture, they can be differentiated by three key characteristics:

The number of layers

The direction of information travel

The number of nodes within each layer

The number of layers

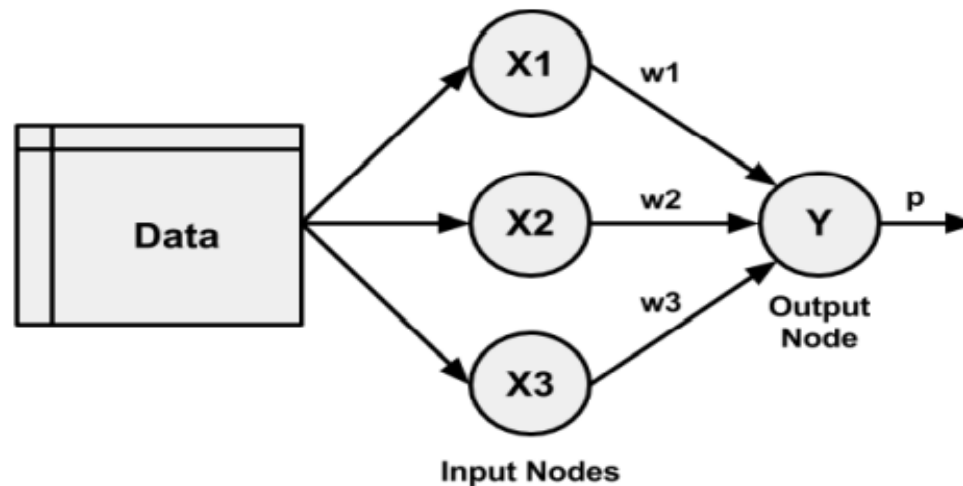
- ▶ The input and output nodes are arranged in groups known as layers.

1) Single-layer network

2) Multi-layer network

Single-layer network

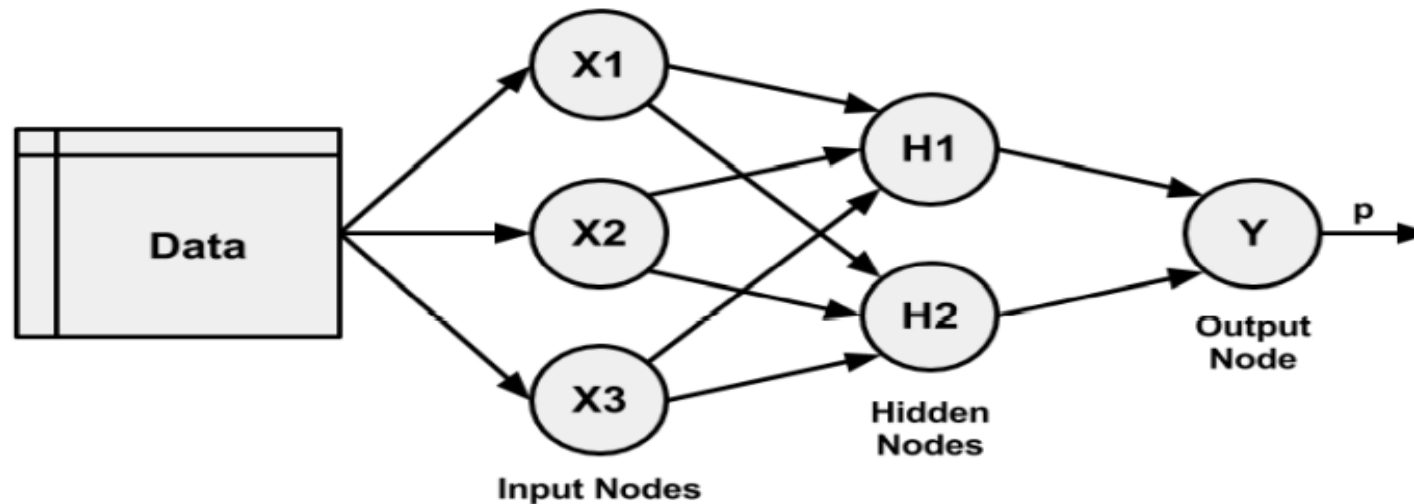
- ▶ Because the input nodes process the incoming data exactly as received, the network has only one set of connection weights (labeled as w_1 , w_2 , and w_3).
- ▶ It is therefore termed a single-layer network.



- ▶ Single-layer networks can be used for basic pattern classification, particularly for patterns that are linearly separable, but more sophisticated networks are required for most learning tasks.

Multi-layer network

- ▶ An obvious way to create more complex networks is by adding additional layers.
- ▶ A multilayer network adds one or more hidden layers that process the signals from the input nodes prior to reaching the output node.



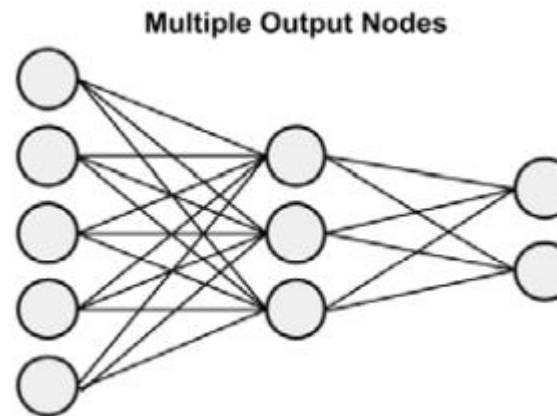
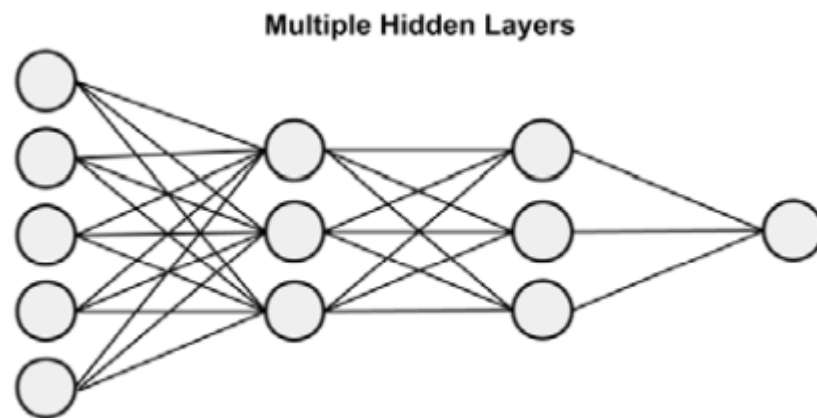
- ▶ Most multilayer networks are fully connected, which means that every node in one layer is connected to every node in the next layer, but this is not required.

The direction of information travel

- ▶ feedforward networks
- ▶ feedback networks (recurrent network)

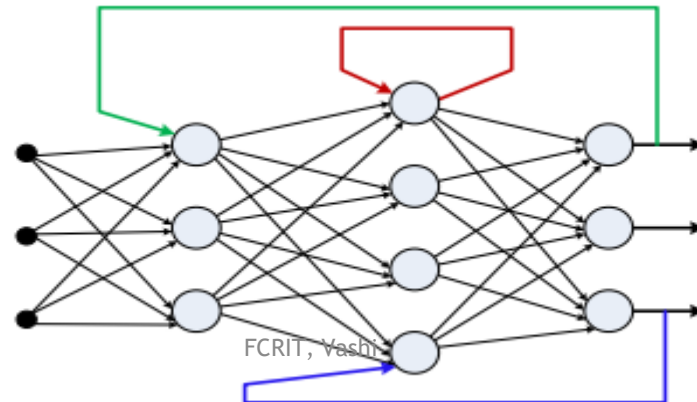
Feedforward Networks

- Networks in which the input signal is fed continuously in one direction from connection-to-connection until reaching the output layer are called feedforward networks.



Feedback Networks

- ▶ A recurrent network (or feedback network) allows signals to travel in both directions using loops.
- ▶ This property, which more closely mirrors how a biological neural network works, allows extremely complex patterns to be learned.
- ▶ The addition of a short term memory increases the power of recurrent networks immensely.
- ▶ Notably, this includes the capability to understand sequences of events over a period of time.
- ▶ This could be used for stock market prediction, speech comprehension, or weather forecasting.



The Training Algorithm

- ▶ The network topology is a blank slate that by itself has not learned anything.
- ▶ Like a newborn child, it must be trained with experience.
- ▶ As the neural network processes the input data, connections between the neurons are strengthened or weakened.
- ▶ *The network's connection weights reflect the patterns observed over time.*
- ▶ Training a neural network by adjusting connection weights is very computationally intensive.
- ▶ The algorithm, which used a strategy of back-propagating errors, is now known simply as backpropagation.

The Training Algorithm

Strengths	Weaknesses
• Can be adapted to classification or numeric prediction problems • Among the most accurate modeling approaches • Makes few assumptions about the data's underlying relationships	• Reputation of being computationally intensive and slow to train, particularly if the network topology is complex • Easy to overfit or underfit training data • Results in a complex black box model that is difficult if not impossible to interpret

The Training Algorithm

- ▶ In its most general form, the backpropagation algorithm iterates through many cycles of two processes.
- ▶ Each iteration of the algorithm is known as an **epoch**.
- ▶ Because the network contains no a priori (existing) knowledge, typically the weights are set randomly prior to beginning.
- ▶ Then, the algorithm cycles through the processes until a stopping criterion is reached.
- ▶ The cycles include:
 - ▶ **A forward phase** in which the neurons are activated in sequence from the input layer to the output layer, applying each neuron's weights and activation function along the way.
 - ▶ Upon reaching the final layer, an output signal is produced.
 - ▶ **A backward phase** in which the network's output signal resulting from the forward phase is compared to the true target value in the training data.
- ▶ The difference between the network's output signal and the true value results in an error that is propagated backwards in the network to modify the connection weights between neurons and reduce future errors.
- ▶ **Over time**, the network uses the information sent backward to reduce the total error of the network.

Learning Rule

- ▶ Yet one question remains:
- ▶ because the relationship between each neuron's inputs and outputs is complex,
- ▶ how does the algorithm determine how much (or whether) a weight should be changed?
- ▶ The answer to this question involves a technique called gradient descent.
- ▶ In a similar process, the backpropagation algorithm uses the derivative of each neuron's activation function to identify the gradient in the direction of each of the incoming weights—hence the importance of having a differentiable activation function.
- ▶ The gradient suggests how steeply the error will be reduced or increased for a change in the weight.
- ▶ **The algorithm will attempt to change the weights that result in the greatest reduction in error by an amount known as the learning rate.**
- ▶ **The greater the learning rate, the faster the algorithm will attempt to descend down the gradients.**

Learning Rules in ANNs

